

KARUNIYA PREMNATH

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EDUCATION

Northeastern University, Boston, Massachusetts Present
Master of Science in Data Science
Related Courses: Linear Algebra for Data Science, Machine Learning, Essentials of Data Science

Saranathan College of Engineering, Tamil Nadu, India 2021 – 2025
Bachelor of Technology in Artificial Intelligence & Data Science
CGPA: 7.94/10

PROFESSIONAL EXPERIENCE

GNANI.AI, India Jun 2023
Software Developer Intern

- Enhanced speech clarity using Audacity through advanced noise reduction and volume normalization.
- Supported AI-based speech processing workflows during live project implementation.
- Improved presentation-quality audio outputs for deployment-ready systems.

INFILABS, India Aug 2023
Software Developer Intern

- Assisted in development and testing of AI-driven modules.
- Collaborated with development team for debugging and deployment support.

PROJECT EXPERIENCE

AI Powered Smart Farming Assistant Apr 2025

- Developed AI-driven agricultural assistant integrating IoT sensors and machine learning models.
- Achieved 85%+ accuracy in crop health prediction using predictive analytics.
- Automated irrigation and pest monitoring using real-time environmental data.
- Built user-friendly dashboard for remote monitoring and device control.

Safe Commute with Crime Hotspots – Aware Route Suggestion Using SVM Apr 2025

- Developed smart routing system recommending safer travel paths using SVM classification.
- Achieved 80%+ accuracy in crime hotspot prediction through model tuning.
- Integrated Google Maps API for real-time route visualization.
- Designed dynamic safety scoring mechanism for improved urban mobility.

Wireless Cylinder Safety Monitoring System Aug 2024

- Designed IoT-based gas leakage detection system with real-time monitoring.
- Implemented wireless data transmission for continuous safety tracking.
- Developed alert notification system for emergency response.

TECHNICAL SKILLS

Programming: Python, SQL, R
Machine Learning: SVM, LSTM, TensorFlow, Keras, Scikit-learn
Data Visualization: Matplotlib, Seaborn
Tools & Concepts: Google Maps API, Audacity, Deep Learning, IoT Integration, Predictive Analytics

CERTIFICATIONS

Executive Post Graduate Certification in Data Science & AI – IIT Roorkee (Intellipaat)
Advance SoC Design for PPA – NIT Trichy (PROBE 2023)
Code Like a Ninja: Cloud Solutions Workshop – Zoho Catalyst